

REVATI NATU

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Education

K. J. Somaiya School of Engineering, Somaiya Vidyavihar University

Oct. 2022 – Jun 2026

B. Tech in Information Technology with Honors in Artificial Intelligence (CGPA: 9.4)

Mumbai, MH

Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Data Science, Data Visualization, Data Analysis, Statistics, Cloud Computing, Software Methodology, Database Management Systems, Operating Systems

Technical Skills

Languages: Python, C++, C, HTML, CSS, JavaScript, SQL, React JS

AI ML Frameworks: TensorFlow, PyTorch, scikit-learn, Hugging Face Transformers, LangChain, PyRadiomics, GeoPandas

Technologies: Pandas, NumPy, OpenCV, Matplotlib, LangChain, Hugging Face, GitHub, Linux

GenAI & LLMs: Prompt Engineering, LLM Fine-tuning, RAG, Gemini API, ElevenLabs, Veo, Magnific

Tools and Skills: Git, GitHub, Linux, VS Code, Jupyter Notebook, Google Colab

Certifications: Generative AI with LLMs, Neural Networks and Deep Learning, Cybersecurity Certification (Google)

Experience

Jio Creative Labs

Jan 2026 – Jun 2026

AI and Creatives Intern

Mumbai, MH

- Built BizIntel, an independent data scraping and intelligence agent using Agentic AI, extending the unified platform with automated business research capabilities and client outreach data for internal workflows.
- Delivered 100+ client-facing AI assets (brand visuals, animated end slates, and multi-character audio) across platforms.

ATRX Mutation Prediction in Glioblastoma - Link

Apr 2026 – Jun 2026

Research Collaborator — Stanford University

Remote

- Designed and implemented an end-to-end explainable ML pipeline for brain tumor genomic profiling from MRI data (Frontiers in Oncology, Q1/Q2), covering 535+ radiomic features, multi-classifier optimization, and SHAP interpretability.
- Achieved 31% sensitivity improvement (0.737 → 0.947) via multimodal MRI feature fusion, validated through bootstrapped CIs and paired DeLong testing across 106 patients.

Research Experience

Multi-Cancer Classification via Deep Learning Techniques

Aug 2025 – Nov 2025

- Developed a deep learning framework for multi-cancer histopathology classification across 65,000+ images; DenseNet201 achieved 99.93% accuracy, outperforming baseline CNN and Vision Transformer architectures.

Forgery Detection System via Inpainting Techniques - Link

Feb 2025 – Apr 2025

- Developed a computer vision system for forgery detection achieving 95.52% accuracy; applied image inpainting on a large public dataset with confidence scores and presented it at an international conference.

Projects

CAPTURED — AI Image Caption Generator

May 2026

- Built a browser-based AI image caption generator with zero backend infrastructure, integrating Gemini 2.5 Flash for holistic scene understanding across 5 caption styles and multiple tones via client-side Gemini API calls.

Research Paper Summarizer and Chatbot

May 2025

- Built a Streamlit-based tool for uploading and querying research papers using Gemini Pro and LangChain, enabling efficient literature reviews with structured metadata extraction, CSV summarization, and smart Q&A from PDFs.

Diabetes Prediction System

Feb 2024

- Evaluated 10+ ML algorithms and developed a Linear Stacking Classifier for diabetes prediction; Led the research from pre-processing to model evaluation and presented at international conference, awarded Best Paper.

Leadership / Extracurricular

Agastya International Foundation (Funded by JPMC)

Oct 2024 – Feb 2025

Engineering Volunteer

Mumbai, MH

- Mentored teams of 5 school students as a technical volunteer, offering hands-on guidance and engineering expertise to overcome technical challenges, complete projects successfully, and boost engagement and confidence in STEM concepts.

National Service Scheme (NSS), SVU

Aug 2023 – Sep 2023

General Secretary

Mumbai, MH

- Directed a 100+ member team as NSS General Secretary, streamlining roles to drive impactful community initiatives and hosting over 5 brainstorming sessions to enhance collaboration, boosting event success rates.